Incremental predictive value of a spectral fragility index for atrial reentry: a pre-registered in-silico evaluation with a clinical recurrence endpoint

Short title. Spectral fragility index for atrial reentry

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Reporting guideline. TRIPOD+AI, for the clinical endpoint.

Scope. Computational (in-silico) study. All ground-truth labels are electrophysiology-simulator inducibility verdicts with a single exception: the pre-registered clinical endpoint, scored against real two-year post-ablation arrhythmia recurrence in the 82 University of Washington patients. No post-operative atrial fibrillation outcome was simulated.

Word count. Main text 2908; abstract 236. Two tables, three figures, 19 references.

Abstract

Background. Algebraic connectivity λ_2 and its closed-form per-edge sensitivity govern stability in coupled excitable networks, motivating a biomarker of atrial reentry, the Spectral Fragility Index (SFI). Whether it adds information beyond established fibrosis and connectivity features was unknown.

Methods. Under a protocol pre-registered before any label was inspected, SFI was added to six competitor features on 182 patient-derived left-atrial meshes from two public cohorts (Roney, $n = 100$; University of Washington, $n = 82$) and up to 100,000 synthetic excitable networks. The in-silico outcome was a simulator's reentry-inducibility verdict; a clinical arm used two-year post-ablation arrhythmia recurrence in the 82 Washington patients. Models were fitted under patient- and shape-family-held-out cross-validation and compared by paired DeLong test. The primary endpoint required grouped $\Delta\text{AUC} \geq 0.05$ with $p < 0.05$.

Results. The primary endpoint was not met. On the combined cohorts (42 inducible of 182), gradient boosting gave $\Delta\text{AUC} = +0.012$ (bootstrap CI -0.017 to $+0.044$; $p = 0.449$), an interval that excludes the 0.05 threshold; synthetic-network ΔAUC converged to $+0.003$. The clinical gate fired ($\Delta\text{AUC} = +0.197$; $p = 0.0061$), but the pre-specified plan identified an artefact of an anti-predictive baseline (AUC 0.250) that nine columns of Gaussian noise reproduced (empirical $p = 0.333$); that endpoint was not met. The estimator was ill-conditioned on real tissue (median $\rho \approx 2422$; validity limit $\rho^* \approx 3$).

Conclusions. SFI added no practically meaningful predictive value for atrial reentry beyond existing features, and its one nominally positive patient-outcome result was a false positive its own pre-registered guards caught. These findings do not support clinical use.

Introduction

Atrial fibrillation is sustained on a remodelled conduction substrate. Interstitial fibrosis slows and disorganises propagation, and the resulting wavelength shortening, source–sink mismatch, and unidirectional block permit reentry. Because that substrate is a network of coupled excitable elements, graph-theoretic and spectral summaries of the atrial conduction network have been proposed as substrate biomarkers. One such summary is algebraic connectivity λ_2, the second-smallest Laplacian eigenvalue [1]. It governs the stability of synchronised states in coupled excitable and oscillator networks through the master-stability function [2] and the synchronizability eigenratio λ_N/λ_2 [3], and it has been fielded as a real-time seizure detector [4]. Its sensitivity to a single conduction weight has a classical closed form, $\partial\lambda_2/\partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$ [5], where φ_2 is the Fiedler vector. Aggregating that sensitivity over a fibrosis-weighted conduction-uncoupling field yields a per-region Spectral Fragility Index (SFI).

Whether such an index carries information beyond the substrate features already in use has not been established. Fibrosis burden, fibrosis spatial heterogeneity, fibrosis patch size, and connectivity measures such as the global minimum cut, the percolation threshold, and λ_2 itself are computed from the same meshes at negligible cost. A spectral index earns a place beside them only if it improves discrimination when added to them, by a margin fixed in advance as the one that would matter, under cross-validation that holds out whole patients and shape families so anatomy cannot leak between training and test folds. Evaluations imposing all of this at once have not been reported.

This study evaluated SFI as a strict add-on to six competitor features, predicting simulator reentry inducibility in 182 patient-derived left-atrial meshes (100 Roney [6], 82 University of Washington [7]) and in up to 100,000 synthetic excitable networks. A pre-registered clinical endpoint followed: two-year post-ablation arrhythmia recurrence in the 82 University of Washington patients. The endpoints, cross-validation scheme, baseline feature set, effect-size threshold ($\Delta\text{AUC} \geq 0.05$), and the accepted null that SFI merely re-encodes fibrosis were frozen before any label was inspected.

Methods

Design and pre-registration. The endpoints, cross-validation scheme, baseline feature set, and the accepted null were frozen and git-tagged before any label was inspected. The null hypothesis that SFI merely re-encodes fibrosis was pre-registered as an accepted, reportable outcome. The section of the pre-registration governing the clinical analysis was frozen before any feature was placed beside the outcome column; that analysis was run once and is reported as it came out.

Reporting guideline. One analysis here is scored against a real patient outcome: prediction of two-year post-ablation arrhythmia recurrence in the 82 University of Washington (UW) patients. That endpoint is reported in accordance with TRIPOD+AI [8]. Adherence is claimed for that endpoint alone. Every other outcome here is an electrophysiology simulator's verdict on reentry

inducibility, which falls outside the statement's remit, though those analyses follow the same frozen protocol.

Cohorts. Two public patient-derived left-atrial mesh deposits were used. The Roney virtual cohort [6] (Zenodo 5801337) contains 100 meshes and all 100 were analysed; fibrosis is a continuous fraction mapped from the shipped image-intensity ratio [9]. The UW/Boyle late-gadolinium-enhancement cohort [7] (Dryad `10.5061/dryad.kkwh70sg0`) contains 164 meshes, being 82 patients imaged before and after ablation; the primary analysis used the 82 pre-ablation meshes only, because post-ablation geometry encodes the delivered treatment. That release ships no intensity ratios (withheld under its institutional review board), so its fibrosis was derived from the per-element tissue tags and is two-valued. No subject was excluded for any other reason, giving 182 analysed meshes with no attrition. Both deposits are de-identified and carry anatomy only, with no age, sex, race, ethnicity, or comorbidity variables. The per-patient two-year recurrence outcomes were supplied on request by the UW cohort's authors, who confirmed they carry no confidentiality restriction. Synthetic excitable networks (up to 100,000 random geometric graphs) were used only for the scale behaviour of the null and for cross-medium transfer.

Conduction graph and index. Nodes were mesh vertices and edges triangle edges, with weight $w_{ij} = [c_\perp + (c_\parallel - c_\perp)a_{ij}^2] \cdot \mathrm{clip}(1 - \phi_{ij}, \varepsilon, 1)$, where a_{ij} is edge–fibre alignment, ϕ_{ij} mean endpoint fibrosis, $c_\parallel = 1.0$, $c_\perp = 0.3$, and $\varepsilon = 0.05$; $L = D - W$ has algebraic connectivity λ_2 with Fiedler vector φ_2. Diffuse fibrosis-weighted uncoupling Δw (mean $\Delta w \approx 0.36$, frozen in the pre-registration) gives, to first order,

$$\mathrm{SFI}(R) \;\approx\; \sum_{(i,j)\in R} \mathbb{E}[\Delta w_{ij}]\,(\varphi_{2,i} - \varphi_{2,j})^2, \qquad \frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2, \tag{1}$$

a classical identity. First-order, basis-independent subspace, and exact-recompute variants were all computed on $\sim$2000-node coarsened graphs.

Comparator features and outcomes. SFI was evaluated strictly as an add-on to six published competitor features: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2 alone. The in-silico outcome was a monodomain Mitchell–Schaeffer [10] inducibility verdict (self-sustained reentry $\geq$ 650 ms after burst pacing at a cycle length of 150 ms from random sites), 42 of 182 meshes inducible; this is a simulator verdict and not clinical atrial fibrillation. Clinically, the outcome was recurrence, defined as at least 30 seconds of documented atrial fibrillation or flutter after a 90-day blanking period and within two years; all 82 patients were followed the full two years, with no censoring, and 48 had events.

Statistical analysis. Two classifiers were used. The first was a standardised, penalised logistic regression whose regularisation was chosen in an inner GroupKFold over $\{0.03, 0.1, 0.3, 1.0, 3.0\}$.

The second was a deliberately untuned, frozen gradient-boosted tree ensemble (120 trees, depth 2, learning rate 0.05, subsample 0.8), whose fixed hyperparameters are logged as a deviation from the pre-registered nested selection. Cross-validation was GroupKFold with the group set to the patient, so no patient appeared in both training and test folds, using $k = \min(5, n_{\text{groups}})$ folds; a naive shuffled-fold arm was reported alongside so that leakage is visible. Out-of-fold predicted probabilities were pooled and compared by the paired DeLong test [11] in its fast midrank form [12]; because pooled scores come from k fitted models rather than two fixed scoring functions, these p values are approximate. Intervals came from at least 10^4 stratified bootstrap resamples. The primary endpoint required *both* grouped $\Delta\text{AUC} \geq 0.05$ and DeLong $p < 0.05$. That effect-size gate was not reused for the clinical arm. With 82 patients and 48 events, power at $\Delta\text{AUC} = 0.05$ is 0.24, so the minimum detectable effect was fixed at $\Delta\text{AUC} \approx 0.10$–$0.15$. The plan also pre-committed that if the baseline came out materially weaker than the source study's published performance, any increment would be reported as confounded by baseline degradation and not as a win for SFI. Testing followed a fixed sequence halting at the first non-rejection. Full methods appear in the Supplementary material.

Results

All 100 Roney meshes and all 82 pre-ablation University of Washington (UW) meshes entered the analysis, 182 in total, with no attrition and no exclusion on any other ground. The electrophysiology simulator returned a reentry-inducible verdict in 34 of 100 Roney subjects (34.0%) and in 8 of 82 UW subjects (9.8%), 42 of 182 combined. The UW rate sits far below the moderate-burst band expected for a cohort of atrial fibrillation patients, and we flag it as anomalous [13–17]. Up to 100,000 synthetic excitable networks supplied the scale arm. Table 1 gives cohort composition, event counts, baseline discrimination, and the incremental results.

The pre-registered primary endpoint, a grouped $\Delta\text{AUC} \geq 0.05$ with DeLong $p < 0.05$ when SFI was added to the six published competitor features, was not met in any cohort or by either classifier. On the combined real cohort the competitor baseline reached an out-of-fold AUC of 0.884 under logistic regression and 0.851 under gradient boosting; adding the SFI block changed discrimination by $\Delta\text{AUC} = -0.007$ (95% CI -0.042 to $+0.029$; $p = 0.70$) and $+0.012$ (95% CI -0.017 to $+0.044$; DeLong $p = 0.449$), respectively. The gradient-boosting point estimate is roughly a quarter of the pre-registered threshold, and its interval excludes that threshold as well as containing zero. On the combined cohort, then, power is not what limits the answer: the effect was measured, and it was small. Cohort-specific increments were -0.021 and $+0.009$ on Roney and $+0.000$ and $+0.035$ on UW. The UW arm rests on eight positives and its intervals ($[-0.073, +0.081]$ and $[-0.095, +0.203]$) reach past the effect-size gate, so that arm settles nothing either way (Figure 1). The pooled row is a single ROC across two cohorts of unequal prevalence, so it carries between-cohort as well as within-cohort separability and does not replace the cohort-specific estimates.

The null held at scale. On 2000 FitzHugh–Nagumo networks the increment was $\leq +0.003$ with

$p \geq 0.12$, and on 500 Kuramoto phase-oscillator networks $\leq +0.0036$ with $p \geq 0.38$. Extending the FitzHugh–Nagumo cohort to 100,000 networks resolved the effect statistically without enlarging it: the subspace variant added $+0.0028$ at $p < 10^{-4}$ once $N \geq 10^4$. Detecting an increment of that size at 80% power would require approximately 4150 cases, and the single-vector variant's $+0.0007$ approximately 23,000. That characterisation was not transported to the real cohorts, a separate regime.

A zero increment admits three readings: the feature is redundant, it carries no information, or it carries information the classifier cannot exploit. Redundancy was therefore measured on its own, without reference to the label. The leading canonical correlation between the nine SFI columns and the six competitors was 0.996, against 0.289 for a permuted-SFI control, so the two blocks are almost collinear in their dominant direction. The per-column picture was less uniform. The total and region-mean summaries were reconstructible from the competitors at $R^2 = 0.970$, and four of the nine columns reached $R^2 \geq 0.60$. The remaining five (the region maximum, the region standard deviation, the top single region, and both edge-level columns) retained most of their variance at $R^2 < 0.25$, well above the permuted control. Redundancy therefore explains part of the null; the rest lies elsewhere. Where the competitors already carry the information, there is nothing left for SFI to add; where unique variance survives, it does not predict the label. The *exact* per-region $\Delta\lambda_2$ variant, which carries no first-order approximation error, also added nothing over the competitor set, so the approximation is not what failed.

Estimator validity was assessed without reference to any label. The single-vector first-order form is trustworthy only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$, the Davis–Kahan ratio [18], is $O(1)$; empirically its relative error is 1% at $\rho = 0.3$ and crosses 10% at $\rho^* \approx 3$. Across the 12 scanned real atrial meshes the median was $\rho \approx 2422$ (interquartile range 1082–7384; full range 817–16554), so every mesh exceeded ρ^* and even the smallest did so by a factor of 272. The spectral gap in the denominator collapses with resolution: measured on the weighted conduction Laplacian the biomarker itself uses, the atrial surface fitted $N^{-1.00}$, within 0.5% of the Weyl 2-manifold exponent [19] of -1.0. Refining the mesh over the same anatomy therefore drives the estimator further out of its valid range (Figure 2). The finding here is numerical: on real tissue the single-vector estimator is invalid and φ_2 is ill-conditioned. Whether the feature ranks cases well is a separate question, settled only against competitors.

The only endpoint scored against patient outcomes was two-year post-ablation arrhythmia recurrence in the 82 UW patients, 48 of whom recurred, with every patient followed for the full two years (Table 2). At $n = 82$ with 48 events, power at $\Delta\text{AUC} = 0.05$ was 0.24. The analysis plan, frozen and tagged before any feature was placed beside the outcome column, therefore fixed the minimum detectable effect at $\Delta\text{AUC} \approx 0.10$–$0.15$ and specified a fixed-sequence stopping rule. It also pre-committed that a materially degraded baseline would be reported as confounding. Under logistic regression the competitor baseline gave an out-of-fold AUC of 0.250, adding SFI gave 0.447, and the increment was $\Delta\text{AUC} = +0.197$ (95% CI $+0.054$ to $+0.334$; DeLong $p = 0.0061$), clearing both arithmetic thresholds. The pre-specified reading is that the increment is an artefact. A baseline

AUC of 0.250 is anti-predictive, and 0.447 remains below chance. All 15 features, competitor and SFI alike, were univariately at chance (0.435–0.547). The gradient-boosting arm, whose baseline sat at 0.516, gave $\Delta\text{AUC} = +0.021$ ($p = 0.738$). Replacing the nine SFI columns with nine columns of standard Gaussian noise and re-running the identical pipeline reproduced the increment: the noise gained +0.137 on average across 20 draws, with 6 of 20 matching or exceeding +0.197 (empirical $p = 0.333$). The mechanism is visible in the folds. Their event rates ranged from 25% to 81%, and mean predicted probability on a held-out fold tracked that fold's training prevalence at $r = +0.995$, so pooling out-of-fold probabilities inverted the ranking; scoring within folds and averaging returned the baseline to 0.345. That primary endpoint was not met, the observed increment is reported as confounded by baseline degradation, and under the stopping rule the ablation-induced ΔSFI test was neither performed nor computed.

Removing any single pre-registered guard manufactured a specific false positive (Figure 3). On a controlled replicated cohort, replacing shape-family grouping with naive random cross-validation inflated AUC from 0.640 to 0.756, an increment of +0.117. Removing the effect-size gate admitted the p-value trap: at $N = 55{,}000$ networks the subspace SFI reached a paired DeLong $p \approx 2 \times 10^{-24}$ over the competitor set while its ΔAUC of +0.003 would require approximately 4150 cases to detect at 80% power.

Discussion

The pre-registered primary endpoint was not met. Added to six published competitor features under leakage-controlled grouping, SFI left the combined real-cohort gradient-boosting increment at $\Delta\text{AUC} = +0.012$, with a bootstrap interval $[-0.017, +0.044]$ that excludes the pre-registered 0.05 gate as well as containing zero (Table 1, Figure 1). On 182 patient-derived anatomies the effect was measured and came out small; power is not the limiting factor. The lower bound of -0.017 still leaves room for an increment of a few thousandths, which only a far larger cohort could resolve. The synthetic arm behaved the same way, the effect converging to $\Delta\text{AUC} \approx +0.003$ across up to 100,000 networks.

How SFI was computed does not explain the null. The exact per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error, also added nothing over the competitor set. Measuring the mechanism directly, without reference to the label, placed it partly in feature redundancy: the SFI block is nearly collinear with the competitor block in its dominant direction, and the two aggregate summaries can be reconstructed from the competitors almost exactly. The account is incomplete, though, because the region maximum, the region spread and the edge-level columns keep most of their variance. Those columns carry information the competitors do not, and it does not predict the label.

Independently of the predictive question, the validity radius $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$ is a label-free precondition that any Fiedler-based fragility marker can be screened against before an outcome is seen. First-order relative error scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$, while real atria sit at

median $\rho \approx 2422$, every mesh past the threshold (Figure 2). The spectral gap collapses as $N^{-1.00}$ on the weighted conduction Laplacian, matching the Weyl 2-manifold exponent to within 0.5%, so finer meshing degrades the single-vector estimator in a predictable direction. The screen is portable to spectral fragility markers used outside cardiology. It bounds estimator accuracy only, and says nothing about how well a feature ranks cases, which must still be tested against competitors.

The methodological result is that ordinary practice would have reported this biomarker as working. Dropping the shape-family grouping inflates AUC by +0.117, and dropping the effect-size gate lets an increment of +0.003 pass at $p \approx 2 \times 10^{-24}$ (Figure 3). Both failure modes occurred in this project's own pipeline. In silico, a ΔAUC of +0.103 at $p = 0.009$ was withdrawn once 38 further released meshes were added. On real patients, the clinical endpoint returned $\Delta\text{AUC} = +0.197$ at DeLong $p = 0.0061$, clearing both of its thresholds. The analysis plan, frozen before the outcomes were joined, identified that increment as an artefact of baseline degradation: the baseline was anti-predictive at AUC 0.250 and reached only 0.447 with SFI added. Nine columns of Gaussian noise then reproduced it, gaining +0.137 on average with 6 of 20 draws matching or exceeding SFI (empirical $p = 0.333$; Table 2), and the fixed-sequence stopping rule halted testing. The guard that caught the false positive was written before the outcome column was opened, not after the result was seen.

Limitations

The in-silico labels are a simulator's reentry-inducibility verdict, not clinical atrial fibrillation; monodomain Mitchell–Schaeffer was substituted for the pre-registered openCARP ground truth and logged as a deviation. That labeller is strongly resolution-sensitive and does not converge in the tested window. The inducibility rate falls from 58% at 1500 nodes to 17% at 3000, then rises to 37.5% at $\sim$50k, against 34.0% at the 2000-node operating resolution, and only 7/24 subjects are consistent across all six tiers. Per-subject label noise is non-differential and biases every reported increment toward zero, so the null is partly a statement about the labeller. By construction the clinical arm is underpowered: with 48/82 events, power at the 0.05 gate is 0.24, the minimum detectable effect is $\Delta\text{AUC} \approx 0.10$ to 0.15, and 80% power at 0.05 would require 371 to 859 patients, so its null is not evidence of no effect. Neither public deposit released any demographic or clinical variable; both ship anatomical fields only. No sociodemographic characterisation, subgroup analysis, or assessment of whether discrimination differs across subgroups is therefore possible, and the representativeness of these patients relative to any ablation population is unknown. There was no patient or public involvement, this being a secondary analysis of de-identified deposits. Each cohort derives from a single originating centre. openCARP was run only as a robustness check and is barred from any endpoint, its calibration gate passing on one cohort and not the other.

Conclusions

A closed-form spectral fragility index did not add practically meaningful reentry-inducibility prediction beyond existing features, at synthetic scale or on 182 patients of real anatomy. The single-vector

estimator is also ill-conditioned on real tissue, and more so as meshes are refined. The clinical endpoint was not met, and it could not have resolved an effect of the size that would matter. The validity radius, which needs no outcome label, can be applied to other Fiedler-based markers before they are tested.

CRediT authorship contribution statement

Mounish Mavuduru: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics statement

No new human or animal data were collected. Analyses use previously published, de-identified, publicly deposited datasets under a documented human-data exemption determination; no institutional review board approval or informed consent was required.

Data availability

Both anatomical cohorts are public, de-identified deposits: the Roney left-atrial virtual cohort (Zenodo 5801337) and the University of Washington/Boyle cohort (Dryad `10.5061/dryad.kkwh70sg0`). Neither deposit contains an outcome label; the two-year recurrence outcomes were withheld from the public download and obtained by direct request to the cohort's authors, who have confirmed in writing that no confidentiality attaches to them. Analysis code, the frozen pre-registration, and all result files are publicly available in the project repository.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used Claude (Anthropic) to assist with drafting, editing, and checking the manuscript text and analysis code. After using this tool, the author

reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Supplementary material

The Supplementary material is the full technical report, containing the complete methods, the substrate provenance audit and its four controls, the independent solver cross-check, the resolution-convergence study, the cross-medium localization results, the power analyses, and the record of every correction made during this work.

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Table 1: Incremental predictive value of the Spectral Fragility Index added to the published competitor feature set. ΔAUC is leakage-controlled (patient-held-out cross-validation); lr, penalised logistic regression; gbt, gradient-boosted trees. The pre-registered endpoint (grouped $\Delta\text{AUC} \geq 0.05$ *and* DeLong $p < 0.05$) is met in no regime, and every interval on real anatomy contains zero. The UW/Boyle rows rest on eight positives and settle nothing either way.

cohort / medium	events/N	model	base AUC	SFI ΔAUC (95% CI); p
Roney (real)	34/100	lr	0.870	-0.021 $[-0.075, +0.033]$; 0.44
		gbt	0.856	$+0.009$ $[-0.029, +0.046]$; 0.62
UW/Boyle (real)	8/82	lr	0.834	$+0.000$ $[-0.073, +0.081]$; 1.00
		gbt	0.748	$+0.035$ $[-0.095, +0.203]$; 0.67
Combined real	42/182	lr	0.884	-0.007 $[-0.042, +0.029]$; 0.70
		gbt	0.851	$+0.012$ $[-0.017, +0.044]$; 0.45
FHN networks	up to 100,000			subspace $+0.0028$; $p < 10^{-4}$ at $N \geq 10^4$
Kuramoto networks	500 networks			$\leq +0.0036$; $p \geq 0.38$

Table 2: Pre-registered clinical endpoint: two-year post-ablation arrhythmia recurrence in 82 University of Washington patients (48 events). The arithmetic gate fired under logistic regression, and the analysis plan frozen before outcomes were joined identifies the increment as an artefact of an anti-predictive baseline: nine columns of Gaussian noise substituted for the nine SFI columns reproduce it. The endpoint is *not* met. †The source study's model uses 89 features including electronic-health-record and clinical risk factors not held here; the comparison establishes only that this baseline is materially weaker, and no claim is made against that model.

analysis	base AUC	+SFI AUC	ΔAUC	95% CI	DeLong p
Logistic regression	0.250	0.447	$+0.197$	$[+0.054, +0.334]$	0.0061
Gradient-boosted trees	0.516	0.537	$+0.021$	$[-0.100, +0.142]$	0.738
Pre-specified guard and post-hoc diagnostics					
Noise control (20 draws)	mean ΔAUC $+0.137$; 6/20 draws $\geq$ observed				empirical 0.333
Fibrosis-only baseline	best AUC 0.603, against the source study's 0.80†				
Within-fold scoring	recovers base AUC $0.250 \rightarrow 0.345$				

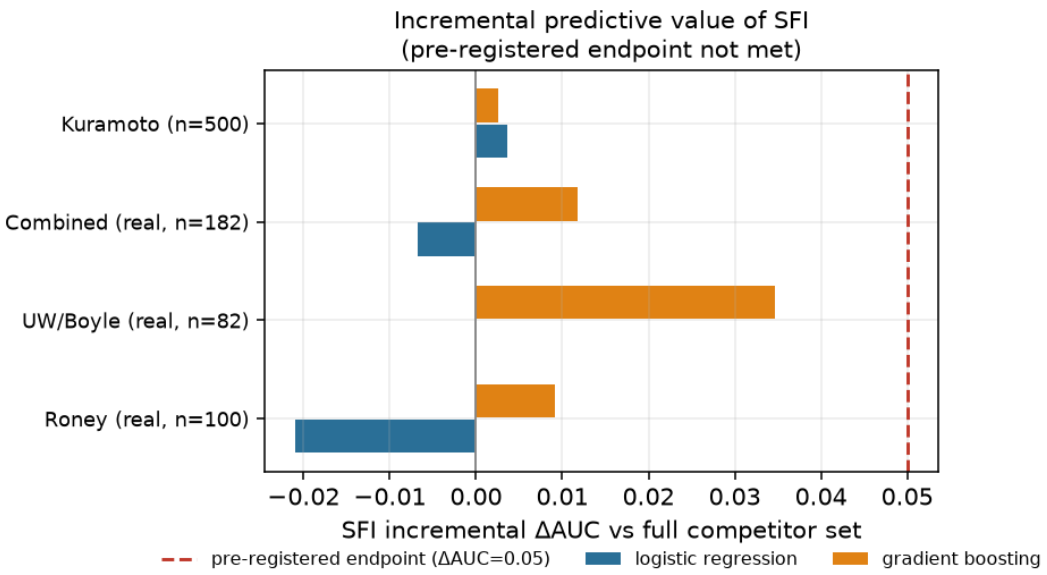


Figure 1: The predictive null. Adding the Spectral Fragility Index to the competitor set does not clear the pre-registered effect-size gate in any cohort; on the combined real cohort the bootstrap interval excludes the gate as well as containing zero.

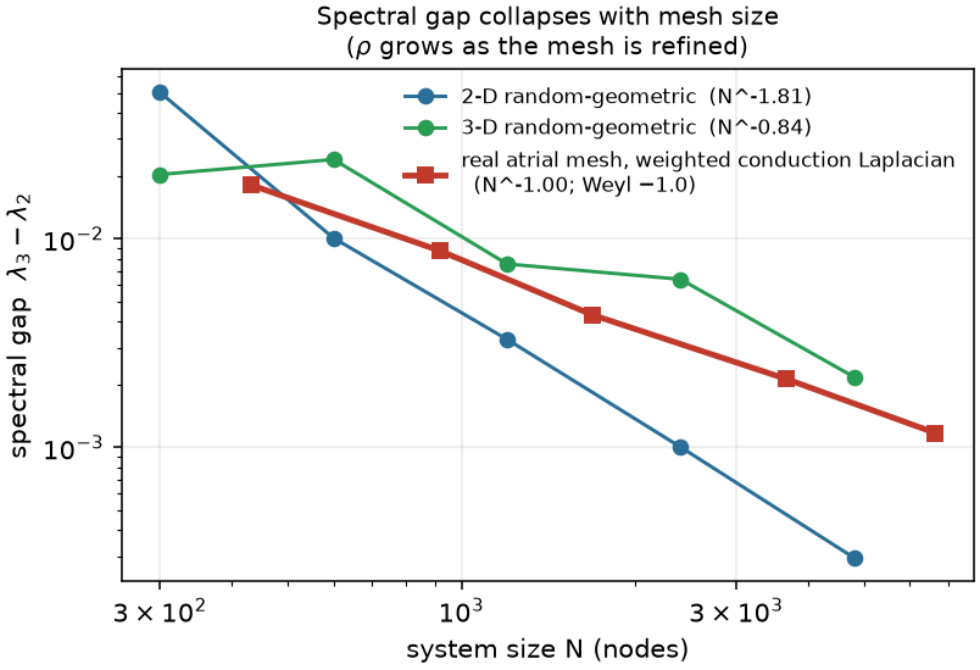


Figure 2: The estimator-validity boundary. The spectral gap collapses as the mesh is refined, at an exponent within 0.5% of the Weyl 2-manifold value, so ρ grows and the first-order estimator degrades *more* on a finer mesh of the same anatomy. The atrial curve is measured on the weighted conduction Laplacian the biomarker itself uses.

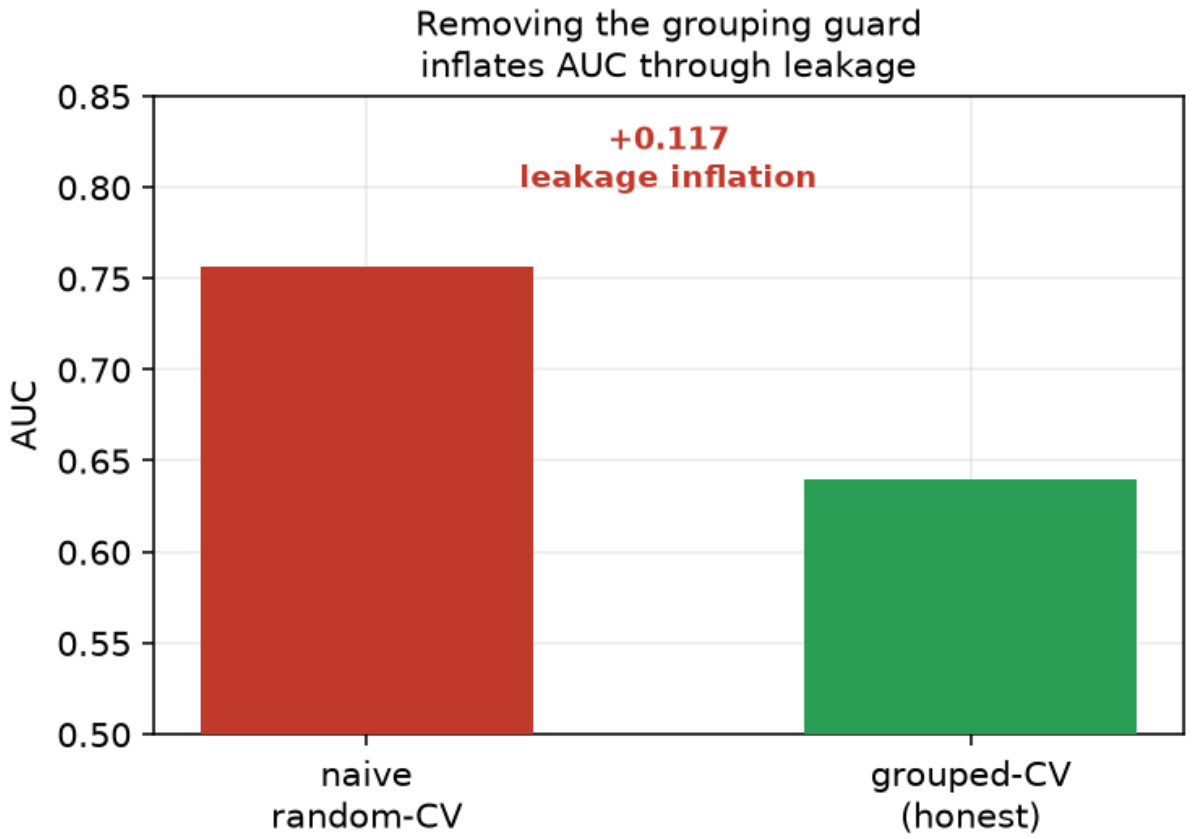


Figure 3: What the pre-registered guards caught. Removing shape-family grouping inflates AUC by +0.117; removing the effect-size gate lets an increment of +0.003 pass at $p = 2 \times 10^{-24}$ on 55,000 networks.